Module - IV

Linear Transformations and applications

Linear transformations, Basic properties, invertible linear transformation, matrices of linear transformations - vector space of linear transformations, change of bases, Similarity

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1 **Defn** Let V and W be vector spaces,
A function $T: V \to W$ is called a linear transformation from V to W if $\forall\, x, y \in V$ and scalar k

(i) $T(x+y) = T(x) + T(y)$

(ii) $T(kx) = kT(x)$

or $T(x+ky) = T(x) + kT(y)$

Ex Let $T: P_1 \to P_2$ defined by
$$T(at+b) = t(at+b)$$
Is T a linear transformation?

Soln Let $(at+b)$ and $(ct+d)$ are vectors in P_1 & k be a scalar

$$T\{(at+b)+(ct+d)\} = t\{(at+b)+(ct+d)\}$$
$$= t(at+b) + t(ct+d)$$
$$= T(at+b) + T(ct+d)$$

$$T[k(at+b)] = t[k(at+b)] = k[t(at+b)]$$
$$= kT(at+b)$$

$\therefore$ T is a linear transformation.

Ex

Let $T: P_1 \to P_2$

$$T[p(t)] = t\,p(t) + t^2$$

is L a linear transformation ?

Sol^n

Let $p(t)$ & $q(t) \in P_1$

k be a scalar.

$$T[p(t) + q(t)] = t[p(t) + q(t)] + t^2$$
$$= t\,p(t) + t\,q(t) + t^2$$

&

$$T[p(t)] + L[q(t)] = [t\,p(t) + t^2] + [t\,q(t) + t^2]$$
$$= t[p(t) + q(t)] + 2t^2$$

$\Rightarrow$ $T[p(t) + q(t)] \neq T[p(t)] + T[q(t)]$

$\Rightarrow$ T is not a linear transformation.

Theorem Let $T: V \longrightarrow W$ be a L.T.

Then

(1) $T(0_V) = 0_W$ where 0_V and 0_W are the zero vectors in V and W.

(2) For any $x_1, x_2, \dots, x_n \in V$ and scalars $k_1, k_2, \dots, k_n$

$$T(k_1x_1 + k_2x_2 + \dots + k_nx_n) = k_1T(x_1) + k_2T(x_2) + \dots + k_nT(x_n)$$

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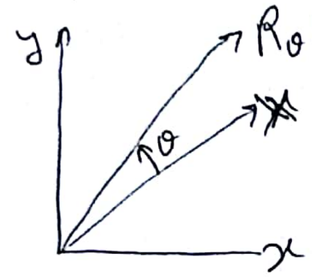
Non-trivial examples of linear transformation

(i) Rotation

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

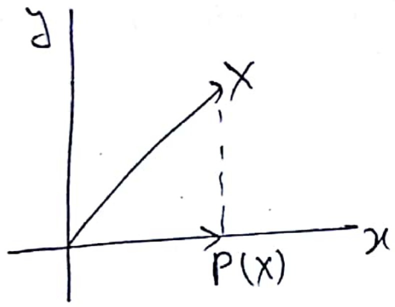
where θ is angle b/w x-axis and a fixed vector in R^2.

It rotates any vector in R^2 through the angle θ.



Rotation by angle θ.

(ii) Projection



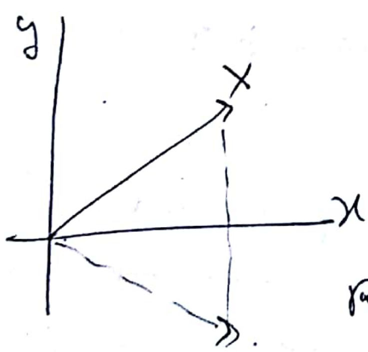
$P : R^2 \to R^2$

For $X \in R^2$

$$P(X) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

Projection on x axis.

(iii) Reflection



$T : R^2 \to R^2$

$$T(x) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$$

Reflection about x-axis.

Defn Let V and W be two vector spaces, and let $T: V \to W$ be a linear transformation from V into W.

(i) $\text{Ker}(T) = \{v \in V : T(v) = 0\} \subseteq V$ is called the kernel of T.

(ii) $\text{Im}(T) = \{T(v) \in W : v \in V\} = T(V) \subseteq W$ is called the image of T.

Theorem Let $T: V \to W$ be a L.T. from a vector space V to a vector space W. Then the kernel Ker(T) and the image Im(T) are subspaces of V & W, respectively.

Proof $\because T(0) = 0$

$\Rightarrow$ Ker(T) & Im(T) are nonempty sets having 0 element.

(1) For any $x, y \in \text{Ker}(T)$ and scalar k.

$$T(x+ky) = T(x) + kT(y) = 0 + k0 = 0$$

$\Rightarrow x + ky \in \text{Ker}(T)$

$\Rightarrow$ Ker(T) is a subspace of V.

(2) If $v, w \in \text{Im}(T)$, then $\exists$ x and $y \in V$ s.t. $T(x) = v$ & $T(y) = w$

For any k.

$$v + kw = T(x) + kT(y) = T(x+ky)$$

$\Rightarrow v + kw \in \text{Im}(T)$

$\Rightarrow$ Im(T) is a subspace of W.

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* $\text{Ker}(A) = N(A)$ and $\text{Im}(A) = C(A)$ for matrix A.

Let $A: \mathbb{R}^n \to \mathbb{R}^m$ a L.T. defined by $m \times n$ matrix A.

$\Rightarrow$ Ker(A) = Solution of the homogeneous system $Ax = 0$

Im(A) = Column space C(A) of matrix A

i.e. $C(A) = \text{Im}(A) \subseteq \mathbb{R}^m$

Theorem Let V and W be vector spaces. Let $\{v_1, v_2, \ldots, v_n\}$ be a basis for V and let $w_1, w_2, \ldots, w_n$ be any vectors in W. Then there exists a unique linear transformation $T: V \to W$ s.t. $T(v_i) = w_i$ for $i = 1, 2, \ldots, n$

Corollary Let V and W be vectors spaces and let $\{v_1, \ldots, v_n\}$ be a basis for V.
If $S, T: V \to W$ are L.T.s and $S(v_i) = T(v_i)$ for $i = 1, 2, \ldots, n$ then $S = T$, i.e. $S(x) = T(x)$ $\forall x \in V$.

Ex Let $T: P_1 \to P_2$ be a L.T.
and $T(t+1) = t^2 - 1$ and $T(t-1) = t^2 + t$
what is $T(7t+3)$?

Soln $\{t+1, t-1\}$ form basis for P_1.
We can write $(7t+3)$ as a l.c. of $(t+1)$ & $(t-1)$

$$7t + 3 = 5(t+1) + 2(t-1)$$

then

$$\begin{aligned} T(7t+3) &= T[5(t+1) + 2(t-1)] \\ &= 5T(t+1) + 2T(t-1) \\ &= 5(t^2-1) + 2(t^2+t) \\ &= 7t^2 + 2t - 5 \end{aligned}$$

6

Ex Let $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

(a) Is T onto?

(b) Find a basis for $\mathrm{Im}\, T$.

(c) Find $\ker T$.

(d) Is T one-to-one?

Onto $\Rightarrow$ If $\mathrm{Im}(T) = W$ in L.T. $T: V \to W$

i.e. $T(V) = W$

For every $w \in W$, there is $v \in V$ s.t. $T(v) = w$.

One-to-one $\Rightarrow$

$T: V \to W$

$V_1 \neq V_2 \Rightarrow L(V_1) \neq L(V_2)$

or $L(V_1) = L(V_2) \Rightarrow V_1 = V_2$

Soln (a) Let $w = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} \in \mathbb{R}^3$

If we can find a vector $v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in $\mathbb{R}^3$

$T(v) = w$?

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$$

RREF
$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & a_1 \\ 0 & 1 & 1 & b_1 - a_1 \\ 0 & 0 & 0 & c_1 - b_1 - a_1 \end{array}\right]$$

Soln will exist only if $c_1 - b_1 - a_1 = 0$

$\Rightarrow$ T is not onto.

(b) basis for $\mathrm{Im}\, T$:

$$\because T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a + c \\ a + b + 2c \\ 2a + b + 3c \end{bmatrix}$$

$$= a\begin{bmatrix}1\\1\\2\end{bmatrix} + b\begin{bmatrix}0\\1\\1\end{bmatrix} + c\begin{bmatrix}1\\2\\3\end{bmatrix}$$

$$\Rightarrow \left\{ \underbrace{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}}_{L.I.}, \begin{bmatrix}1\\2\\3\end{bmatrix} \right\} \text{ spans } T.$$

First two vectors are L.I. 3^{rd} vector is sum of first two

$$\Rightarrow \left\{ \begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix} \right\} \text{ form basis for } \operatorname{Im} T.$$

(c) Ker T.

$T(v) = 0$.

Null space.

$$\begin{bmatrix}1 & 0 & 1\\1 & 1 & 2\\2 & 1 & 3\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix} = 0$$

$$\sim \begin{bmatrix}1 & 0 & 1\\0 & 1 & 1\\0 & 0 & 0\end{bmatrix} \Rightarrow \begin{aligned} c &= t\\ b &= -t\\ a &= -t \end{aligned}$$

$$= t\begin{bmatrix}-1\\-1\\1\end{bmatrix}, \quad t \in \mathbb{R}$$

(d) $\because \operatorname{Ker} T \neq \{0_{\mathbb{R}^3}\}$

$\Rightarrow$ T is not one-to-one.

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Ex. (Linear extension of a transformation defined on a basis)

Let $w_1 = (1,0)$, $w_2 = (2,-1)$, $w_3 = (4,3) \in \mathbb{R}^2$.

(1) Let $\alpha = \{e_1, e_2, e_3\}$ be standard basis for $\mathbb{R}^3$.

$T : \mathbb{R}^3 \to \mathbb{R}^2$ be the L.T. defined by

$T(e_1) = w_1$, $T(e_2) = w_2$, $T(e_3) = w_3$

Find formula for $T(x_1, x_2, x_3)$ and compute $T(2,-3,5)$

Sol^n Let $x = (x_1, x_2, x_3) \in \mathbb{R}^3$

$\Rightarrow$ It can be written as l.c. of basis elements

$x = x_1 e_1 + x_2 e_2 + x_3 e_3 \in \mathbb{R}^3$

$$\therefore \quad T(x) = T(x_1 e_1 + x_2 e_2 + x_3 e_3)$$
$$= x_1 T(e_1) + x_2 T(e_2) + x_3 T(e_3)$$
$$= x_1 w_1 + x_2 w_2 + x_3 w_3$$
$$= x_1(1,0) + x_2(2,-1) + x_3(4,3)$$
$$= (x_1 + 2x_2 + 4x_3, -x_2 + 3x_3)$$

$$\therefore \quad T(2,-3,5) = (2-6+20, 3+15) = (16, 18)$$

(2) Let $\beta = (V_1, V_2, V_3)$ be another basis

$V_1 = (1,1,1)$, $V_2 = (1,1,0)$, $V_3 = (1,0,0)$

and $T : \mathbb{R}^3 \to \mathbb{R}^2$ be L.T.

$T(V_1) = w_1$, $T(V_2) = w_2$, $T(V_3) = w_3$

Find $T(x_1, x_2, x_3)$ and compute $T(2,-3,5)$?

Soln Let $(x_1, x_2, x_3) \in \mathbb{R}^3$

$$(x_1, x_2, x_3) = c_1(1,1,1) + c_2(1,1,0) + c_3(1,0,0)$$

compare both sides and solve for c_1, c_2, c_3

$$c_1 = x_3$$
$$c_2 = x_2 - x_3$$
$$c_3 = x_1 - x_2$$

$$\therefore (x_1, x_2, x_3) = x_3(1,1,1) + (x_2 - x_3)(1,1,0) + (x_1 - x_2)(1,0,0)$$
$$= x_3 V_1 + (x_2 - x_3) V_2 + (x_1 - x_2) V_3$$

$$T(x_1, x_2, x_3) = x_3 T(V_1) + (x_2 - x_3) T(V_2) + (x_1 - x_2) T(V_3)$$
$$= x_3(1,0) + (x_2 - x_3)\cancel{(2,1)}(2,-1) + (x_1 - x_2)(4,3)$$
$$= (4x_1 - 2x_2 - x_3,\ 3x_1 - 4x_2 + x_3)$$

$$\therefore T(2,-3,5) = (8+6-5,\ 6+12+5)$$
$$= (9, 23)$$

Invertible linear transformation

Let $f: X \to Y$

if $\exists\ g: Y \to X$ s.t. $g \circ f = id$

and $f \circ g = id$

Then g is called the inverse function of f

$g = f^{-1}$

* For a given sets to be invertible is that is one-to-one and onto.

$f: X \to Y$ is one-to-one (injective)

if $f(u) = f(v)$ in $Y \Rightarrow u = v$ in X

and $f: X \to Y$ is onto (or surjective)

if for each $y \in Y\ \exists\ x$ in X

s.t. $f(x) = y$.

Bijective $\Rightarrow$ one-to-one + onto

* If $T: V \to W$ and $S: W \to Z$ are L.T.

Then $(S \circ T)(v) = S(T(v))$ also a L.T.

<u>Lemma</u>. Let V & W be vector space. If $T: V \to W$ is invertible LT, then its inverse $T^{-1}: W \to V$ is also linear

<u>Proof</u> Let $w_1, w_2 \in W$ & k any scalar

$\because T$ is invertible $\Rightarrow \exists\ v_1$ & v_2 in V

s.t. $T(v_1) = w_1$ & $T(v_2) = w_2$

Then $T^{-1}(w_1 + kw_2) = T^{-1}(T(v_1) + kT(v_2))$

$= \cancel{T^{-1}(T v_1 + k} = \cancel{T^{-1}(T(v_1+v_2)}$

$= T^{-1}(T(v_1 + kv_2))$

$= v_1 + kv_2$

$= T^{-1}(w_1) + kT^{-1}(w_2)$

Def^n Isomorphism

A L.T. $T : V \to W$ is called isomorphism if it is invertible (or one-to-one and onto).
$\Rightarrow$ V and W are isomorphic to each other.

* A vector space can be identified with another if they are isomorphic to each other.
  Ex The vector space $P_n(R)$ can be identified with the $(n+1)$-space R^{n+1}.

* If T is an isomorphism $\Rightarrow T^{-1}$ also an isomorphism

* A L.T. $A : R^n \to R^n$ defined by square matrix $A_{n \times n}$
  A will be isomorphism iff A is invertible
  i.e. $\text{Rank}(A) = n$

* A L.T. $T : V \to W$ is one-to-one iff $\text{Ker}(T) = \{0\}$
  & If $V = W$, then T is one-to-one iff T is onto.

Imp **Th^m** Two vector spaces V and W are isomorphic iff $\dim V = \dim W$.

Proof Let $T : V \to W$ be isomorphism
Let $\{V_1, V_2, \dots, V_n\}$ be a basis for V.
$\Rightarrow$ If $\{T(V_1), T(V_2) \dots T(V_n)\}$ is a basis for W
$\Rightarrow \dim W = \dim V = n$

To prove $\{T(V_1) \dots T(V_n)\}$ ~~we have to~~ basis we have to check the linear independence & span. To

(1) Linear Independent.
If T is one-to-one, then
$$0 = c_1 T(V_1) + c_2 T(V_2) + \dots + c_n T(V_n)$$
$$= T(c_1 V_1 + c_2 V_2 + \dots + c_n V_n)$$

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$\Rightarrow \quad 0 = c_1 V_1 + c_2 V_2 + \cdots + c_n V_1$

$\because$ the V_i's are L.I.

$\therefore \quad c_i = 0 \quad \forall\, i = 1, 2, \ldots, n$

(2) Spans W

$\because$ T is onto

$\Rightarrow$ For any $y \in W \quad \exists \; x \in V$

s.t. $\quad T(x) = y$

Let $\quad x = \sum_{i=1}^{n} a_i V_i$

$$y = T(x) = T(a_1 V_1 + a_2 V_2 + \cdots + a_n V_n)$$

$$= a_1 T(V_1) + a_2 T(V_2) + \cdots + a_n T(V_n)$$

$\Rightarrow$ Any $y \in W$ can be written as as linear combinatn of vectors of W. i.e. it spans W.

Conversely Let $\dim V = \dim W = n$

$\Rightarrow$ The basis of V and W

$(V_1, V_2, \ldots, V_n)$ and $(W_1, W_2, \ldots, W_n)$ both will have n elements

$\exists$ a L.T.

$T: V \to W$ and $S: W \to V$

s.t. $T(V_i) = W_i$ and $S(W_i) = V_i, \; i = 1, 2, \ldots, n$

$\Rightarrow (S \circ T)(V_i) = V_i$ and $(T \circ S)(W_i) = W_i$

$\Rightarrow (S \circ T)$ and $(T \circ S)$ are identity transformation on V & W.

$\Rightarrow$ T and S are isomorphism

Corollary Let V and W be vector spaces

(1) If $\dim V = n$ then V is isomorphic to the n-space R^n.

(2) If $\dim V = \dim W$, any bijective function from a basis for V to a basis for W can be extended to an isomorphism from V to W.

* Let V be a vector space of dim n with an ordered basis $\alpha = \{v_1, v_2, \ldots, v_n\}$ and let $\beta = \{e_1, e_2, \ldots, e_n\}$ be the standard ordered basis for $\mathbb{R}^n$.

Then isomorphism $\phi : V \longrightarrow \mathbb{R}^n$

defined by $\phi(v_i) = e_i$ is called the natural isomorphism w.r.t. the basis α.

$\Rightarrow$ a vector in V can be identified with a column vector in $\mathbb{R}^n$.

For any $x = \sum_{i=1}^{n} a_i v_i \in V$

$$\phi(x) = \sum_{i=1}^{n} a_i \, \phi(v_i) = \sum_{i=1}^{n} a_i e_i = (a_1, \ldots, a_n) = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n$$

i.e. coordinate vector of x w.r.t the basis α.

denoted by $[x]_\alpha$

$\Rightarrow$ $[v_i]_\alpha = e_i$

Ex. Let $T: P_2 \to P_2$ be the L.T.

$$L(at^2+bt+c) = (a+2b)t + (b+c)$$

(a) Is $(-4t^2+2t-2)$ in Ker T ?

(b) Is t^2+2t+1 in range T ?

(c) Find a basis for ker T.

(d) Is T one-to-one ?

(e) Find a basis for rang T

(f) Is T onto ?

Sol^n (a) $T(-4t^2+2t-2) = (-4+4)t + (2-2)$
$= 0$

$\Rightarrow (-4t^2+2t-2)$ is in Ker T.

(b) (t^2+2t+1) is in rang T if :

$\exists$ a vector at^2+bt+c in P_2

s.t. $L(at^2+bt+c) = t^2+2t+1$

$\Rightarrow (a+2b)t + (b+c) = t^2+2t+1$

$\Rightarrow$ $0 = 1$
$a+2b = 2$. Inconsist syst^n
$b+c = 1$

$\Rightarrow (t^2+2t+1)$ is not in range L.

(c) $T(at^2+bt+c) = 0$

$\Rightarrow (a+2b)t + (b+c) = 0$

$\Rightarrow$ $a+2b = 0$
$b+c = 0$

Sol^n space $= \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right\}$

$\Rightarrow$ Basis for ker T $= \{2t^2 - t + 1\}$

(d) $\because$ $\ker T \neq \{0\}$

$\Rightarrow$ T is not one-to-one

(e) Basis for range T

$\because (a+2b)t + (b+c)$

$\because$ t and 1 span range T, which are L.I.

$\therefore$ $\{t, 1\}$ form basis for rang T.

(f) $\because$ $\dim(\text{rang } T) = 2$

while $\dim(P_2) = 3$

$\Rightarrow$ rang $T \neq P_2$

$\Rightarrow$ T is not onto.

Matrices of Linear Transformation

A L.T. of a vector space into another can be represented by matrix via the natural isomorphism between n dim vector space and n-space $\mathbb{R}^n$.

Product of $A_{m\times n}$ and an $n\times 1$ matrix is a L.T. from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Let $T: V \to W$ be a L.T. from n-dim V to m-dim W

Let $\alpha = \{v_1, \dots, v_n\}$ is basis for V

$\beta = \{w_1, \dots, w_m\}$ is basis for W

$\because$ The L.T. T of any element is completely determined by its values on a basis,

$$T(v_1) = a_{11}w_1 + a_{12}w_2 + \dots + a_{m1}w_m$$

$$T(v_2) = a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m$$

$$\vdots$$

$$T(v_n) = a_{1n}w_1 + a_{2n}w_2 + \dots + a_{mn}w_m$$

For any vector $x \in V$

$$[T(x)]_\beta = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = A[x]_\alpha$$

i.e. for any $x \in V$ the coordinate vector $[T(x)]_\beta$ of $T(x)$ in W is just a product of a fixed matrix A and the coordinate vector $[x]_\alpha$ of x.

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = \left[[T(V_1)]_\beta \cdots [T(V_n)]_\beta \right]$$

A is the matrix whose column vectors are just the coordinate vectors $[T(V_j)]_\beta$ of $T(V_j)$ w.r.t. the basis β.

<u>Defn</u> The matrix A is called the associated matrix for T (or the matrix representation of T) w.r.t. the ordered bases α and β, and is denoted by

$$A = [T]_\alpha^\beta$$

when $V = W$ and $\alpha = \beta$ then simply $[T]_\alpha$ for $[T]_\alpha^\alpha$

<u>Theorem</u> Let $T: V \to W$ be a L.T. from an n-dim vector space V to an m-dim vector space W. For fixed ordered bases $\alpha = \{V_1, V_2, \ldots, V_n\}$ for V and β for W, there corresponds a unique associated $m \times n$ matrix $[T]_\alpha^\beta$ for T such that for any vector $x \in V$ the coordinate vector $[T(x)]_\beta$ of $T(x)$ w.r.t. β is given as a matrix product of the associated matrix $[T]_\alpha^\beta$ for T and the coordinate vector $[X]_\alpha$.

i.e $$[T(X)]_\beta = [T]_\alpha^\beta [X]_\alpha$$

The associated matrix $[T]_\alpha^\beta$ is given as

$$[T]_\alpha^\beta = \left[[T(V_1)]_\beta \; [T(V_2)]_\beta \cdots [T(V_n)]_\beta \right]$$

Procedure to find the matrix of L.T.

$T: V \to W$ w.r.t. basis $\alpha = \{V_1, V_2, \dots V_n\}$ & $\beta = \{W_1, \dots, W_m\}$ of V & W respectively

Step. I Compute $T(V_i)$. for $i = 1, \dots, n$

Step-II Find coordinate vector $[T(V_i)]_\beta$ i.e. express $T(V_i)$ as a l.c. of vector of β

Step-III The matrix A of T w.r.t bases α and β is formed by choosing $[T(V_i)]_\beta$ as the ith column of A.

Ex Let $T: \mathbb{R}^3 \to \mathbb{R}^2$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \end{bmatrix}$$

Let $\alpha = \{V_1, V_2, V_3\}$, & $\beta = \{W_1, W_2\}$ bases for $\mathbb{R}^3$ and $\mathbb{R}^2$.

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T(V_1) = T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T(V_2) = T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$T(V_3) = T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Now $[T(V_1)]_\beta = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $[T(V_2)]_\beta = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $[T(V_3)]_\beta = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

$\therefore$ The matrix of linear transformation

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Ex $T : R^3 \to R^2$ Bases for R^3 $\alpha = \{V_1, V_2, V_3\}$

Bases for R^2 $\beta = \{W_1, W_2\}$

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \end{bmatrix} \qquad V_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, V_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, V_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, W_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$T(V_1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

$$= a_1 W_1 + a_2 W_2 = a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$T(V_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = b_1 W_1 + b_2 W_2 = b_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$T(V_3) = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = c_1 W_1 + c_2 W_2 = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

i.e. three linear systems in two unknowns in each can be solved once

$$\left[\begin{array}{cc|c|c|c} 1 & -1 & 1 & 1 & 2 \\ 2 & 1 & -1 & 0 & 0 \end{array}\right]$$

RREF

$$\left[\begin{array}{cc|c|c|c} 1 & 0 & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & -1 & -\frac{2}{3} & -4/3 \end{array}\right]$$

$\therefore$ matrix of L.T. w.r.t. bases α and β is

$$A = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ -1 & -\frac{2}{3} & -\frac{4}{3} \end{bmatrix}$$

Ex $L: P_1 \longrightarrow P_2$

$L[p(t)] = t\,p(t)$

(a) Find matrix of L w.r.t base $S = \{t, 1\}$ and $T = \{t^2, t-1, t+1\}$ for P_1 & P_2, respectively.

(b) If $p(t) = 3t-2$, compute $L[p(t)]$ using the matrix.

Sol^n (a) $L(t) = 1(t^2) + 0(t-1) + 0(t+1) = t^2$

$$\Rightarrow [L(t)]_T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$L(1) = t = 0(t^2) + \frac{1}{2}(t-1) + \frac{1}{2}(t+1)$$

$$[L(1)]_T = \begin{bmatrix} 0 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$\therefore$ Matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{bmatrix}$

(b) $$[L[p(t)]]_T = A[p(t)]_S = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}$$

$$L(3t-2) = 3t^2 + (-1)(t-1) + (-1)(t+1)$$
$$= 3t^2 - 2t$$

Ex Let $L: P_1 \longrightarrow P_2$

$L[p(t)] = t\,p(t)$

(a) Find the matrix of transformation w.r.t. bases $S = \{t, 1\}$ and $T = \{t^2, t, 1\}$ for P_1 and P_2, respectively

(b) If $p(t) = 3t - 2$, compute $L[p(t)]$ using matrix.

Soln (a) $L(t) = t^2 + 0 \times t + 0 \times 1 \Rightarrow [L(t)]_T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$L(1) = 0 \times t^2 + t + 0 \times 1 \Rightarrow [L(1)]_T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\therefore$ matrix of transformation from S to T

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

(b) $L[p(t)] = t\,p(t) = t(3t-2) = 3t^2 - 2t$

Given $p(t) = 3t + (-2)1$

$\Rightarrow [p(t)]_S = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$\Rightarrow [L[p(t)]]_T = A\,[p(t)]_S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$

$\Rightarrow L[p(t)] = 3t^2 - 2t$

* If the matrix of L.T. is an invertible $n \times n$ square matrix, then the column vectors $c_1, c_2, \ldots, c_n$ form another basis γ for $\mathbb{R}^n$.

* Let V and W be vector spaces with bases α & β. $T: V \to W$ be a L.T. with matrix representation $[T]_\alpha^\beta = A$.

Then $\ker(T)$ and $\operatorname{Im}(T)$ are isomorphic to the null space $N(A)$ and the column space $C(A)$, respectively.

i.e. if $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ with standard basis

$\ker(T) = N(A)$ and $\operatorname{Im}(T) = C(A)$

$$\boxed{\dim \ker(T) + \dim \operatorname{Im}(T) = \dim V}$$

Ex Let $T: \mathbb{R}^3 \to \mathbb{R}^3$

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+c \\ a+b \\ b-c \end{bmatrix}$$

For basis of $\ker T$

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Soln space of the homogeneous system

$$a + c = 0$$
$$a + b = 0$$
$$b - c = 0$$

Basis of $\ker T = \left\{\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}\right\} \Rightarrow \dim(\ker T) = 1$

Basis for $\operatorname{Im}(T)$.

$\because$ every vector in range T is of form $\begin{pmatrix} a+c \\ a+b \\ b-c \end{pmatrix}$

i.e. $a\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Basis of $\operatorname{Im}(T) = \left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right\} \Rightarrow \dim(\ker T) + \dim(\operatorname{Im} T) = 1 + 2 = 3 = \dim \mathbb{R}^3$

Vector Spaces of Linear Transformation

Let V and W be two V.S. of dimensions n and m. Let $\mathcal{L}(V;W)$ denote the set of all linear transformations from V to W, i.e.

$$\mathcal{L}(V;W) = \{T : T \text{ is a linear transformation from } V \text{ to } W\}$$

Defn For any two L.T. S and T in $\mathcal{L}(V,W)$ and $\lambda \in \mathbb{R}$, we define the sum $S+T$ and the scalar multiplication λS by

$$(S+T)(v) = S(v) + T(v) \quad \&$$

$$(\lambda S)(v) = \lambda(S(v)) \qquad \forall\ v \in V$$

Clearly $\mathcal{L}(V;W)$ forms a vector space.

Defn Let $T : V \to W$ be a L.T. with bases α and β respectively. Then the associated matrix $[T]_\alpha^\beta$ of T w.r.t. these bases is unique.

$\therefore$ the function

$$\phi : \mathcal{L}(V;W) \longrightarrow M_{m\times n}(\mathbb{R}) \text{ is}$$

defined by

$$\phi(T) = [T]_\alpha^\beta \in M_{m\times n}(\mathbb{R})$$

for $T \in \mathcal{L}(V,W)$ is well defined.

Lemma The function $\phi : \mathcal{L}(V,W) \longrightarrow M_{m\times n}(\mathbb{R})$ is a one-to-one correspondence between $\mathcal{L}(V;W)$ and $M_{m\times n}(\mathbb{R})$.

Lemma. Let V and W be V.S. with ordered bases α and β respectively and let $S, T : V \to W$ be linear. Then we have

$$[S+T]_\alpha^\beta = [S]_\alpha^\beta + [T]_\alpha^\beta$$

and $$[kS]_\alpha^\beta = k[S]_\alpha^\beta$$

Theorem For vector spaces V of dimension n and W of dimension m, the vector space $L(V; W)$ of all linear transformations from V to W is isomorphic to the vector space $M_{m\times n}(\mathbb{R})$ of all $m\times n$ matrices, and

$$\dim L(V; W) = \dim M_{m\times n}(\mathbb{R}) = mn = \dim V \dim W$$

Theorem Let V, W and Z be V.S. ~~of~~ with ordered bases α, β and γ respectively.
Suppose $S : V \to W$ and $T : W \to Z$ are L.T.

Then $$[T\circ S]_\alpha^\gamma = [T]_\beta^\gamma [S]_\alpha^\beta$$

Proof. Let $\alpha = \{v_1, v_2, \ldots v_n\}$

$\beta = \{w_1, w_2, \ldots w_m\}$

$\gamma = \{z_1, z_2, \ldots z_l\}$

Let $[T]_\beta^\gamma = [a_{ij}]$

and $[S]_\alpha^\beta = [b_{pq}]$

Then, for $1 \le i \le n$

$$(T \circ S)(v_i) = T(S(v_i))$$

$$= T\left(\sum_{k=1}^{m} b_{ki} w_k\right) = \sum_{k=1}^{m} b_{ki} T(w_k)$$

$$= \sum_{k=1}^{m} b_{ki} \left(\sum_{j=1}^{l} a_{jk} z_j\right) = \sum_{j=1}^{l} \left(\sum_{k=1}^{m} a_{jk} b_{ki}\right) z_j.$$

It shows that $[T \circ S]_\alpha^\gamma = [T]_\beta^\gamma [S]_\alpha^\beta$.

Ex. Let α be the standard basis for $\mathbb{R}^3$ and let $S, T : \mathbb{R}^3 \to \mathbb{R}^3$ be two linear transformations given by

$S(e_1) = (2, 2, 1)$, $S(e_2) = (0, 1, 2)$, $S(e_3) = (-1, 2, 1)$

$T(e_1) = (1, 0, 1)$, $T(e_2) = (0, 1, 1)$, $T(e_3) = (1, 1, 2)$

compute $[S+T]_\alpha$, $[2T-S]_\alpha$, and $[T \circ S]_\alpha$

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Theorem Let V and W be vector spaces with ordered bases α and β, respectively and let $T: V \to W$ be an isomorphism. Then

$$[T^{-1}]_{\beta}^{\alpha} = ([T]_{\alpha}^{\beta})^{-1}$$

Proof. Since T is invertible

$$\dim V = \dim W$$

and the matrices $[T]_{\alpha}^{\beta}$ and $[T^{-1}]_{\beta}^{\alpha}$ are square and of the same size.

$$[T]_{\alpha}^{\beta} [T^{-1}]_{\beta}^{\alpha} = [T \circ T^{-1}]_{\beta} = [id]_{\beta}$$

is the identity matrix.

$$\therefore \quad [T^{-1}]_{\beta}^{\alpha} = ([T]_{\alpha}^{\beta})^{-1}$$

Q For the vector spaces $P_1(R)$ and R^2, choose the bases $\alpha = \{1, x\}$ for $P_1(R)$ and $\beta = \{e_1, e_2\}$ for R^2, respectively. Let $T: P_1(R) \to R^2$ be the L.T. defined

$$T(a + bx) = (a, a+b)$$

(1) Show that T is invertible

(2) Find $[T]_{\alpha}^{\beta}$ and $[T^{-1}]_{\beta}^{\alpha}$

Sol^n $T: P_1 \to R^2$

$$T(a+bx) = (a, a+b)$$

Let $(u, v) \in R^2$

Then $T^{-1}(u, v) = a + bx$

From given transformation $(u, v) = (a, a+b)$

$\Rightarrow u = a \Rightarrow a = u$

$v = a + b \Rightarrow b = v - u$

$\Rightarrow T^{-1}(u, v) = u + (v-u)x$

$T^{-1}(1,0) = 1 + (-1)x = 1 \cdot 1 + (-1)x$

$T^{-1}(0,1) = 0 + x = 0 \cdot 1 + 1 \cdot x$

$$[T^{-1}]_{\beta}^{\alpha} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Change of bases

$[x]_\alpha$ → coordinate vector of x w.r.t. basis α.

$[x]_\beta$ → coordinate vector of x w.r.t. basis β

in $\mathbb{R}^n$ space.

Whats the relation between $[x]_\alpha$ and $[x]_\beta$?

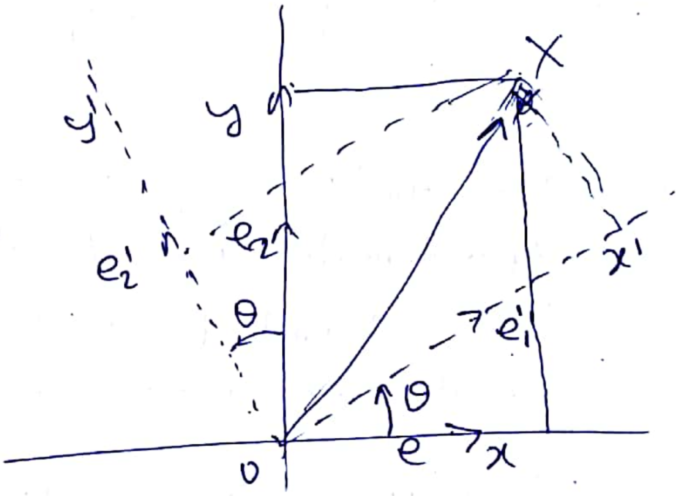
* Let $X = (x, y) \in \mathbb{R}^2$, Standard basis $\alpha = \{e_1, e_2\}$

$$[X]_\alpha = \begin{bmatrix} x \\ y \end{bmatrix}, \qquad X = x e_1 + y e_2$$

Let $\beta = \{e_1', e_2'\}$ another basis obtained by rotating α counterclockwise through an angle θ.

$$X = x' e_1' + y' e_2'$$

$$[X]_\beta = \begin{bmatrix} x' \\ y' \end{bmatrix}$$



The expression of the vectors in β w.r.t. α

$$e_1' = \cos\theta \, e_1 + \sin\theta \, e_2$$
$$e_2' = -\sin\theta \, e_1 + \cos\theta \, e_2$$

So $[e_1']_\alpha = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$, $[e_2']_\alpha = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$

$$\therefore \; X = x' e_1' + y' e_2' = (x'\cos\theta - y'\sin\theta) e_1 + (x'\sin\theta + y'\cos\theta) e_2$$
$$= x e_1 + y e_2$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

or $\quad [X]_\alpha = [id]_\beta^\alpha \, [X]_\beta$

where $[Id]_\beta^\alpha = \left[[e_1']_\alpha \,\vdots\, [e_2']_\alpha \right] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

& $[Id]_\alpha^\beta = \left([Id]_\beta^\alpha\right)^{-1}$

so $[x]_\beta = [Id]_\alpha^\beta [x]_\alpha$

In general

Let $\alpha = \{v_1, v_2, \ldots, v_n\}$

& $\beta = \{w_1, w_2, \ldots, w_n\}$ be two ordered basis for V.

Then $[Id]_\beta^\alpha = \left[[w_1]_\alpha \,\vdots\, [w_2]_\alpha \,\vdots\, \cdots \,\vdots\, [w_n]_\alpha \right]$ from β to α.

or $[Id]_\alpha^\beta = \left[[v_1]_\beta \,\vdots\, [v_2]_\beta \,\vdots\, \cdots \,\vdots\, [v_n]_\beta \right]$ from α to β

Defn The matrix representation $[id]_\beta^\alpha$ of the identity transformation $id : V \to V$ w.r.t. any two ordered bases α and β is called the basis-change matrix or the coordinate-change matrix from β to α.

$\because$ identity transformation is invertible

$$[x]_\beta = [id]_\alpha^\beta [x]_\alpha = \left([id]_\beta^\alpha\right)^{-1} [x]_\alpha$$

Ex Find the basis-change matrix from a basis α to another basis β for the $\mathbb{R}^3$ space.

$$\alpha = \{(1,0,1),\ (1,1,0),\ (0,1,1)\} = \{V_1, V_2, V_3\}$$

& $$\beta = \{(2,3,1),\ (1,2,0),\ (2,0,3)\} = \{W_1, W_2, W_3\}$$

Soln

$$\because \quad [X]_\beta = [id]_\alpha^\beta [X]_\alpha$$

where $[id]_\alpha^\beta = \left[[V_1]_\beta \vdots [V_2]_\beta \vdots \cdots \vdots [V_n]_\beta \right]$

Express V_1 as a linear combinatn. of basis element of β.

i.e. $$V_1 = c_1 W_1 + c_2 W_2 + c_3 W_3$$

$$(1,0,1) = c_1(2,3,1) + c_2(1,2,0) + c_3(2,0,3)$$

$$\Rightarrow \left.\begin{aligned} 2c_1 + c_2 + 2c_3 &= 1 \\ 3c_1 + 2c_2 &= 0 \\ c_1 + 3c_3 &= 1 \end{aligned}\right\} \Rightarrow \begin{aligned} c_1 &= -2 \\ c_2 &= 3 \\ c_3 &= 1 \end{aligned}$$

Similarly

$$V_2 = d_1 W_1 + d_2 W_2 + d_3 W_3$$

$$(1,1,0) = d_1(2,3,1) + d_2(1,2,0) + d_3(2,0,3)$$

$$\left.\begin{aligned} 2d_1 + d_2 + 2d_3 &= 1 \\ 3d_1 + 2d_2 &= 1 \\ d_1 + 3d_3 &= 0 \end{aligned}\right\} \quad \begin{aligned} d_1 &= -3 \\ d_2 &= 5 \\ d_3 &= 1 \end{aligned}$$

& $$V_3 = f_1 W_1 + f_2 W_2 + f_3 W_3$$

$$(0,1,1) = f_1(2,3,1) + f_2(1,2,0) + f_3(2,0,3)$$

$$\left.\begin{aligned} 2f_1 + f_2 + 2f_3 &= 0 \\ 3f_1 + 2f_2 &= 1 \\ f_1 + 3f_3 &= 1 \end{aligned}\right\} \Rightarrow \begin{aligned} f_1 &= 7 \\ f_2 &= -10 \\ f_3 &= -2 \end{aligned}$$

$$\Rightarrow [Id]_{\alpha}^{\beta} = \begin{bmatrix} -2 & -3 & 7 \\ 3 & 5 & -10 \\ 1 & 1 & -2 \end{bmatrix}$$